Aswanth Bindu Ajith

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LinkedIn

1PROFESSIONAL SUMMARY

Highly motivated interdisciplinary M.Sc. candidate combining a Mechanical Engineering foundation with advanced expertise in data science, machine learning, and scientific computing. Passionate about applying ML models and data-driven decision-making to real industrial manufacturing processes to optimize process parameters, cycle times, and product quality. Backed by practical Agile software development experience and proficiency in Python and PyTorch. Passionate about applying quantum computing, scientific simulation to solve complex engineering challenges. Seeking to contribute to Max Planck Institute (Munich Quantum Valley / IMPRS-QST)'s innovative projects.

2TECHNICAL SKILLS

Python (Advanced),
Scientific Computing,
Scikit-Learn,
SciPy,
PyTorch,
Pandas,
Optimization Methods,
Numerical Analysis

3PROFESSIONAL EXPERIENCE

Working Student Software Developer *Germany*
No Border (E-commerce) Feb 2025 Dec 2025

- Contributed to software development projects within an Agile environment, delivering practical solutions while significantly enhancing Python programming skills
- Collaborated effectively within multidisciplinary teams to implement frontend features, demonstrating high adaptability and attention to detail
- Supported comprehensive testing and debugging workflows, refining a structured approach to solving challenging technical problems
- Utilized Git, GitHub, and Linux for version control and development workflows

4EDUCATION

M.Sc. High Performance Computing and Quantum Computing 2024Present
Deggendorf Institute of Technology (DIT), Germany Current Average: 2.6
B.Tech Mechanical Engineering 20192023
APJ Abdul Kalam Technological University, India CGPA: 6.48

5KEY PROJECTS

Large-Scale Computational Simulation Data Analysis
Developed a Python-based computational simulation to process complex datasets and model interactions among approximately 30,000 system entities. Utilized NumPy and Pandas to extract trends, evaluate data, and identify hidden relationships. Transformed complex numerical data into practical insights and structured scientific reports.

Thermal Analysis of Battery Modules
Conducted extensive engineering simulations to evaluate process behavior and thermal dynamics in high-dimensional physical systems. Applied mechanical engineering problem-solving to understand impact of operational parameters on product quality.

Computing Architecture Optimization Methods
Applied mathematical optimization methods and scientific computing workflows to evaluate computational

efficiency of modern architectures. Developed structured analytical approaches to identify computational bottlenecks.

6RESEARCH INTERESTS

High Performance Computing,
Quantum Computing,
Parallel and Distributed Systems

7LANGUAGES

English (Professional Working Proficiency),
Malayalam (Native),
German (A1 (Beginner))